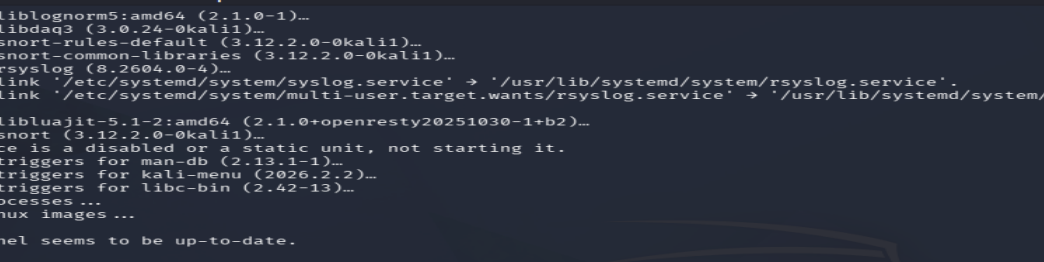


CodeAlpha Internship - Task 4: Network Intrusion Detection System (IDS)

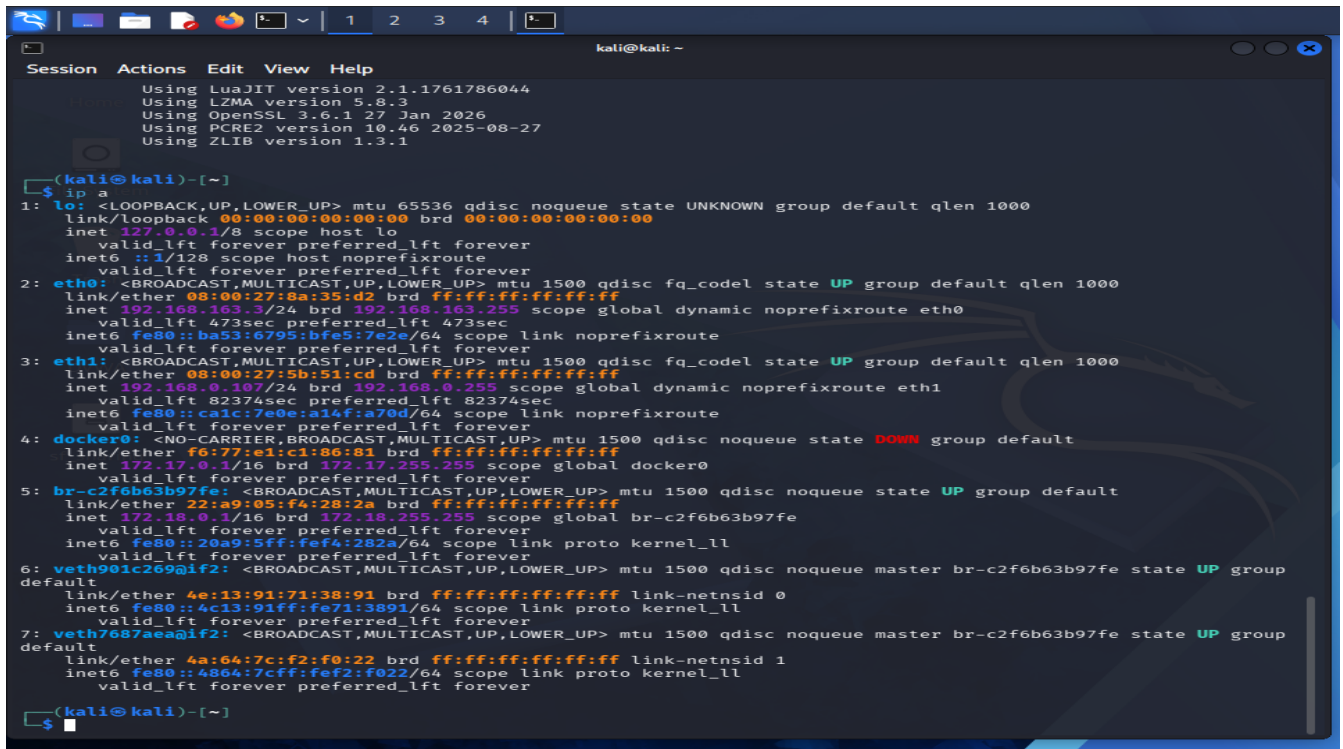
Objective: Install and configure Snort IDS on Kali Linux, create a custom detection rule, generate traffic, and verify alerts.

Figure 1: Snort Installation Verification



```
kali@kali: ~  
Session Actions Edit View Help  
Setting up liblognorm5:amd64 (2.1.0-1)...  
Setting up libdaq3 (3.0.24-0kali1)...  
Setting up snort-rules-default (3.12.2.0-0kali1)...  
Setting up snort-common-libraries (3.12.2.0-0kali1)...  
Setting up rsyslog (8.2604.0-4)...  
Created symlink '/etc/systemd/system/syslog.service' → '/usr/lib/systemd/system/rsyslog.service'.  
Created symlink '/etc/systemd/system/multi-user.target.wants/rsyslog.service' → '/usr/lib/systemd/system/rsyslog.service'.  
Setting up liblua5.1-2:amd64 (2.1.0+openresty20251030-1+b2)...  
Setting up snort (3.12.2.0-0kali1)...  
snort.service is a disabled or a static unit, not starting it.  
Processing triggers for man-db (2.13.1-1)...  
Processing triggers for kali-menu (2026.2.2)...  
Processing triggers for libc-bin (2.42-13)...  
Scanning processes ...  
Scanning linux images ...  
  
Running kernel seems to be up-to-date.  
  
No services need to be restarted.  
  
No containers need to be restarted.  
  
No user sessions are running outdated binaries.  
  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
  
(kali@kali)-[~]  
$  
  
(kali@kali)-[~]  
$ snort -V  
-*) Snort++ <*-  
Version 3.12.2.0  
By Martin Roesch & The Snort Team  
http://snort.org/contact#team  
Copyright (C) 2014-2025 Cisco and/or its affiliates. All rights reserved.  
Copyright (C) 1998-2013 Sourcefire, Inc., et al.  
Using DAQ version 3.0.24  
Using libpcap version 1.10.6 (64-bit time_t, with TPACKET_V3)  
Using LuaJIT version 2.1.1761786044  
Using LZMA version 5.8.3  
Using OpenSSL 3.6.1 27 Jan 2026  
Using PCRE2 version 10.46 2025-08-27  
Using ZLIB version 1.3.1  
  
(kali@kali)-[~]  
$
```

Figure 2: Network Interface Identification

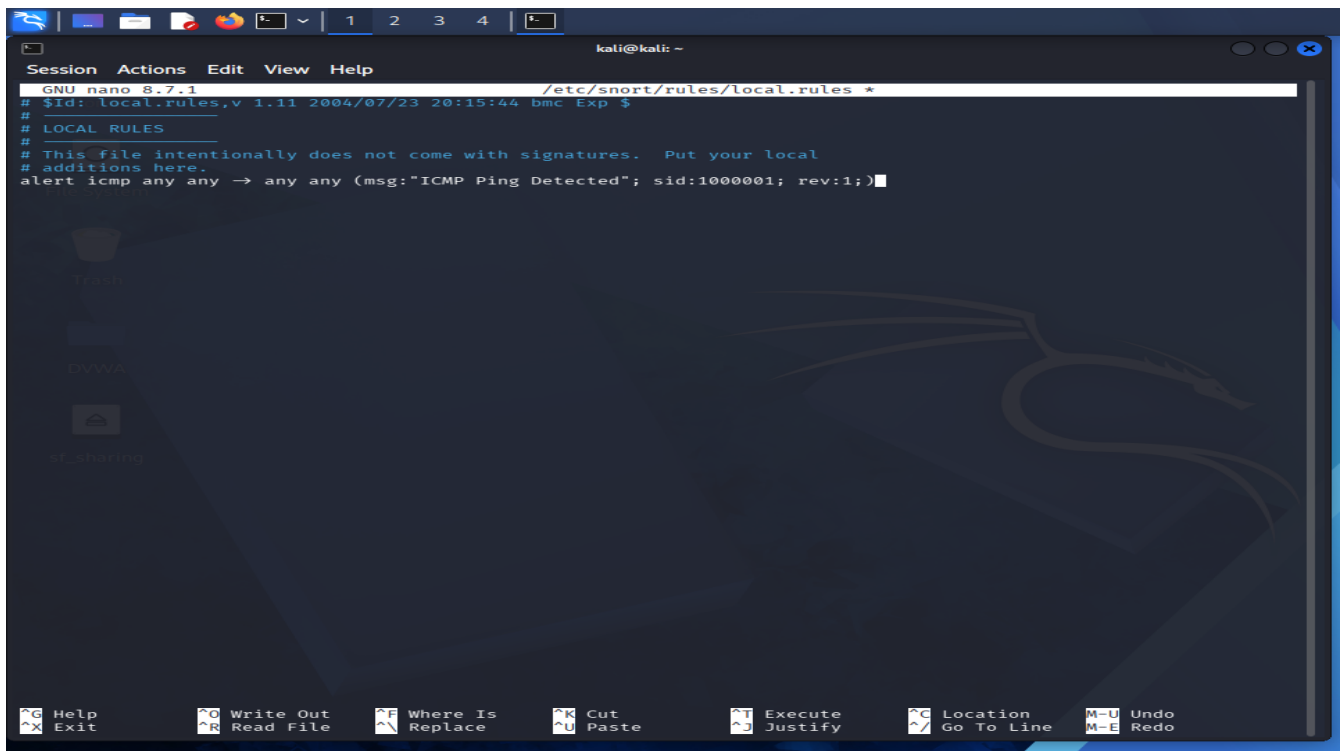


```
Session Actions Edit View Help
Using LuaJIT version 2.1.1761786044
Using LZMA version 5.8.3
Using OpenSSL 3.6.1 27 Jan 2026
Using PCRE2 version 10.46 2025-08-27
Using ZLIB version 1.3.1

(kali@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:8a:35:d2 brd ff:ff:ff:ff:ff:ff
    inet 192.168.163.3/24 brd 192.168.163.255 scope global dynamic noprefixroute eth0
        valid_lft 473sec preferred_lft 473sec
    inet6 fe80::ba53:6795:bfe5:7e2e/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:5b:81:cd brd ff:ff:ff:ff:ff:ff
    inet 192.168.0.107/24 brd 192.168.0.255 scope global dynamic noprefixroute eth1
        valid_lft 82374sec preferred_lft 82374sec
    inet6 fe80::ca1c:7e0e:a14f:a70d/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
4: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether f6:77:e1:c1:86:81 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
5: br-c2f6b63b97fe: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 22:a9:05:f4:28:2a brd ff:ff:ff:ff:ff:ff
    inet 172.18.0.1/16 brd 172.18.255.255 scope global br-c2f6b63b97fe
        valid_lft forever preferred_lft forever
    inet6 fe80::20a9:5ff:fef4:282a/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
6: veth901c269a1f2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-c2f6b63b97fe state UP group default
    link/ether 4e:13:91:71:38:91 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet6 fe80::4c13:91ff:fe71:3891/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
7: veth7687aeag1f2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master br-c2f6b63b97fe state UP group default
    link/ether 4a:64:7c:f2:f0:22 brd ff:ff:ff:ff:ff:ff link-netnsid 1
    inet6 fe80::4864:7cff:fef2:f022/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever

(kali@kali)-[~]
$
```

Figure 3: Custom Rule Configuration



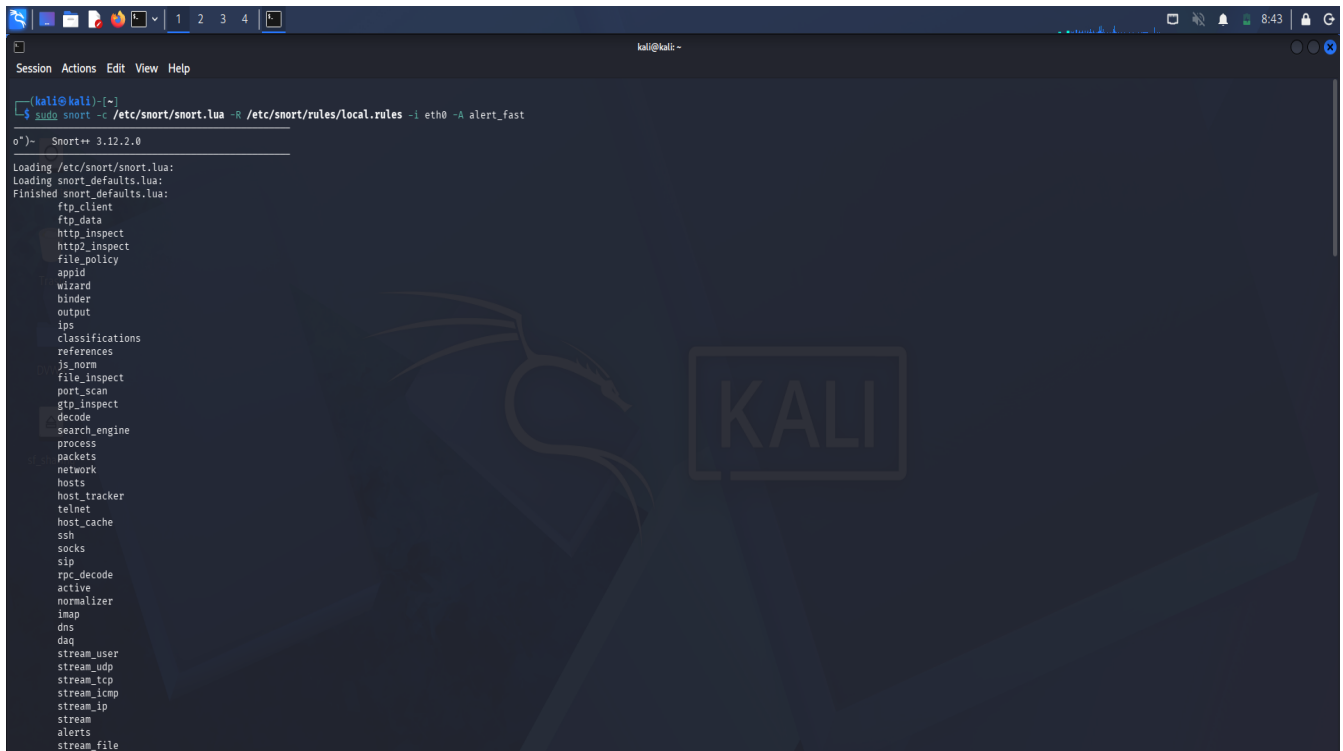
The image shows a Kali Linux desktop environment with a terminal window open. The terminal is running the nano text editor, editing the file `/etc/snort/rules/local.rules`. The editor's status bar at the top indicates "GNU nano 8.7.1" and the file path. The menu bar includes "Session", "Actions", "Edit", "View", and "Help". The terminal content shows the following text:

```
GNU nano 8.7.1 /etc/snort/rules/local.rules *
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
#
# LOCAL RULES
#
# This file intentionally does not come with signatures.  Put your local
# additions here.
alert icmp any any -> any any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)
```

The bottom of the terminal window displays a row of keyboard shortcuts for nano editor functions:

- Help Exit
- Write Out Read File
- Where Is Replace
- Cut Paste
- Execute Justify
- Location Go To Line
- Undo Redo

Figure 4: Snort Monitoring Started



The screenshot shows a Kali Linux terminal window with the title bar 'kali@kali: ~'. The terminal output displays the command to start Snort and the subsequent loading of rules and plugins. A large, faint watermark of a dragon's head and the word 'KALI' is visible in the background of the terminal.

```
(kali@kali)~  
$ sudo snort -c /etc/snort/snort.lua -R /etc/snort/rules/local.rules -i eth0 -A alert_fast  
o*)- Snort++ 3.12.2.0  
Loading /etc/snort/snort.lua:  
Loading snort_defaults.lua:  
Finished snort_defaults.lua:  
ftp_client  
ftp_data  
http_inspect  
http2_inspect  
file_policy  
appid  
wizard  
binder  
output  
ips  
classifications  
references  
js_norm  
file_inspect  
port_scan  
gtp_inspect  
decode  
search_engine  
process  
packets  
network  
hosts  
host_tracker  
telnet  
host_cache  
ssh  
socks  
sip  
rpc_decode  
active  
normalizer  
imap  
dns  
daq  
stream_user  
stream_udp  
stream_tcp  
stream_icmp  
stream_ip  
stream  
alerts  
stream_file
```

Figure 5: ICMP Traffic Generation

```
kali@kali: ~  
Session Actions Edit View Help  
kali@kali ~  
$ ping 8.8.8.8  
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.  
64 bytes from 8.8.8.8: icmp_seq=1 ttl=118 time=32.3 ms  
64 bytes from 8.8.8.8: icmp_seq=2 ttl=118 time=30.8 ms  
64 bytes from 8.8.8.8: icmp_seq=3 ttl=118 time=28.4 ms  
64 bytes from 8.8.8.8: icmp_seq=4 ttl=118 time=30.2 ms  
64 bytes from 8.8.8.8: icmp_seq=5 ttl=118 time=30.2 ms  
64 bytes from 8.8.8.8: icmp_seq=6 ttl=118 time=29.9 ms  
64 bytes from 8.8.8.8: icmp_seq=7 ttl=118 time=29.5 ms  
64 bytes from 8.8.8.8: icmp_seq=8 ttl=118 time=30.6 ms  
64 bytes from 8.8.8.8: icmp_seq=9 ttl=118 time=30.0 ms  
64 bytes from 8.8.8.8: icmp_seq=10 ttl=118 time=28.9 ms  
64 bytes from 8.8.8.8: icmp_seq=11 ttl=118 time=38.8 ms  
64 bytes from 8.8.8.8: icmp_seq=12 ttl=118 time=31.7 ms  
64 bytes from 8.8.8.8: icmp_seq=13 ttl=118 time=29.8 ms  
64 bytes from 8.8.8.8: icmp_seq=14 ttl=118 time=29.6 ms  
64 bytes from 8.8.8.8: icmp_seq=15 ttl=118 time=53.4 ms  
64 bytes from 8.8.8.8: icmp_seq=16 ttl=118 time=28.3 ms  
64 bytes from 8.8.8.8: icmp_seq=17 ttl=118 time=31.0 ms  
^C  
--- 8.8.8.8 ping statistics ---  
17 packets transmitted, 17 received, 0% packet loss, time 16189ms  
rtt min/avg/max/mdev = 28.330/31.980/53.441/5.830 ms  
kali@kali ~  
$
```

Figure 6: Snort Alert Detection

```
kali@kali:~$ sudo tcpdump -i eth1 -s 0 -w - 'icmp[0] > 0'
Session Actions Edit View Help
06/21-08:49:46.489338 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] [ICMP] 192.168.0.107 → 8.8.8.8
06/21-08:49:46.489351 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] [ICMP] 8.8.8.8 → 192.168.0.107
06/21-08:49:47.461546 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] [ICMP] 192.168.0.107 → 8.8.8.8
06/21-08:49:47.490420 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] [ICMP] 8.8.8.8 → 192.168.0.107
06/21-08:49:48.468833 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] [ICMP] 192.168.0.107 → 8.8.8.8
06/21-08:49:48.497550 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] [ICMP] 8.8.8.8 → 192.168.0.107
06/21-08:49:49.470421 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] [ICMP] 192.168.0.107 → 8.8.8.8
06/21-08:49:49.500697 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] [ICMP] 8.8.8.8 → 192.168.0.107
06/21-08:49:49.472227 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] [ICMP] 192.168.0.107 → 8.8.8.8
06/21-08:49:50.499511 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] [ICMP] 8.8.8.8 → 192.168.0.107
06/21-08:49:51.473632 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] [ICMP] 192.168.0.107 → 8.8.8.8
06/21-08:49:51.503982 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] [ICMP] 8.8.8.8 → 192.168.0.107
06/21-08:49:52.475135 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] [ICMP] 192.168.0.107 → 8.8.8.8
06/21-08:49:52.510885 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] [ICMP] 8.8.8.8 → 192.168.0.107
06/21-08:49:53.477663 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] [ICMP] 192.168.0.107 → 8.8.8.8
06/21-08:49:53.508677 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] [ICMP] 8.8.8.8 → 192.168.0.107
06/21-08:49:54.479674 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] [ICMP] 192.168.0.107 → 8.8.8.8
06/21-08:49:54.509688 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] [ICMP] 8.8.8.8 → 192.168.0.107
06/21-08:49:55.526873 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] [ICMP] 192.168.0.107 → 8.8.8.8
06/21-08:49:55.558180 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] [ICMP] 8.8.8.8 → 192.168.0.107
06/21-08:49:56.527922 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] [ICMP] 192.168.0.107 → 8.8.8.8
06/21-08:49:56.557535 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] [ICMP] 8.8.8.8 → 192.168.0.107
06/21-08:49:57.531941 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] [ICMP] 192.168.0.107 → 8.8.8.8
06/21-08:49:57.560690 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] [ICMP] 8.8.8.8 → 192.168.0.107
06/21-08:49:58.534720 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] [ICMP] 192.168.0.107 → 8.8.8.8
06/21-08:49:58.564865 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] [ICMP] 8.8.8.8 → 192.168.0.107
06/21-08:49:59.541941 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] [ICMP] 192.168.0.107 → 8.8.8.8
06/21-08:49:59.574737 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] [ICMP] 8.8.8.8 → 192.168.0.107
06/21-08:50:00.544160 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] [ICMP] 192.168.0.107 → 8.8.8.8
06/21-08:50:00.585284 ** [1:1000001:1] "ICMP Ping Detected" ** [Priority: 0] [ICMP] 8.8.8.8 → 192.168.0.107
-- [0] eth1

Packet Statistics
daq
received: 55
analyzed: 55
allow: 55
rx_bytes: 6996

codec
total: 55 (100.000%)
arp: 10 (18.182%)
eth1: 55 (100.000%)
icmp4: 30 (54.545%)
ipv4: 45 (81.818%)
udp: 15 (27.273%)

Module Statistics
ac_full
```